

Etiological profile of hearing loss amongst Lithuanian pediatric cochlear implant users.

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Congenital sensorineural hearing loss is a heterogeneous disorder; its etiological profile varies between populations. Pathogenic variants of GJB2 gene are the major cause of non-syndromic hearing loss. Congenital cytomegalovirus infection (cCMV) is the most important prenatal etiological factor causing hearing loss and other disorders. Perinatal events, syndromes, postnatal infections or traumas are less common. Causes of the remaining one third of hearing loss cases are unknown. To determine the etiological profile of hearing loss in pediatric cochlear implant users in Lithuanian population. The data of 122 children (70 male/52 female; aged 7.6 ± 3.3 years) cochlear implant users were analysed. Medical records of all children recruited in Santaros Clinics (Vilnius, Lithuania) were analysed to identify prenatal, perinatal, or postnatal risk factors based on the adapted list proposed by the Joint Committee of Infant Hearing. Genetic counselling and testing according to the scheme were performed to 101 children. DNA of 117 children was extracted from the DBS on Guthrie cards and CMV DNA detected using real time PCR. Non-syndromic hearing loss was diagnosed in 65 cases (53.3%), 58 of which were GJB2 gene-associated; syndromic hearing loss was diagnosed to 8 children (6.6%). Perinatal (prematurity, low birth weight, hypoxia, hyperbilirubinemia, sepsis, ototoxicity, and meningitis) and postnatal (meningitis) risk factors were associated with hearing loss in 16 (13.1%) and 4 (3.3%) study participants respectively. CMV DNA was detected in 12 samples (9.8%). The cause of hearing loss remained unknown only for 17 (13.9%) children. The major cause of HL in the current study was GJB2 gene alterations. Only 14% of the cohort had congenital hearing loss of unknown origin.

Review of environmental factors affecting hearing.

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The major nongenetic causes of sensorineural hearing loss are exposure to noise, aging, ototoxic drugs, viral and bacterial infections, and interactions between these factors. Regarding exposure to continuous noise, the data base from laboratory and field studies indicates that a risk of hearing loss is present when noise levels exceed 75-80 dBA. As noise level, duration and number of exposures increase so does risk. The data base for other forms of noise (intermittent, impact) is not as established. Risk of hearing loss due to impulse noise increases as the peak SPL exceeds 145-155 dB and as the duration of the impulse, the number of impulses and the number of exposures increase. High-level acoustic impulses can cause severe, permanent hearing loss. Interaction between some steady-state noises and some acoustic impulses can be synergistic, producing extensive injuries to the organ of Corti. Noise can also interact synergistically with some aminoglycoside antibiotics to produce severe injuries in the inner ear. These antibiotics are also capable of producing hearing loss and indeed may do so in up to 55% of the one million persons who receive aminoglycoside antibiotics during the course of treatment for tuberculosis or severe gram-negative infections. Bacterial and viral infections may also produce mild to severe hearing loss. With the development of rubella vaccine and Rhogam, cytomegalovirus may have become the most common cause of congenital deafness. Aging is also a major cause of hearing loss. Exposure to occupational and environmental noise, certain diseases and life styles (diet, stress, drugs) may interact with the specific effects of aging. The result is moderate to severe hearing loss in a majority of older persons.

Cytomegalovirus infection in pregnancy.

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CMV is the most common cause of intrauterine infection in this country, affecting approximately 1% of newborns. As such, CMV infection is a leading cause of deafness and an important contributor to learning disabilities. The virus is transmitted through close contact with infected secretions as well as transplacentally, through sexual contact, and through blood transfusion and organ transplantation. CMV has a high prevalence in the population; antibody to the virus can be detected in 50% to 85% of individuals. Like herpes simplex, the virus can be reactivated and cause recurrent infection. Congenital disease occurs in about 30% to 40% of women with primary infection. Ten percent of infants with congenital infection will have clinical disease at birth. Of these, as many as 30% die. Ninety percent of survivors have sequelae such as deafness, mental retardation, chorioretinitis, and motor deficits. The other 90% of infants with congenital infection are asymptomatic at birth, but 5% to 17% develop long-term sequelae. Congenital disease also occurs with recurrent maternal infection in about 0.2% to 1% of immune women. Congenital infection due to recurrent disease is far less severe, and less than 10% of affected infants have long-term sequelae (Fig 3). The diagnosis of congenital infection can be established using a combination of ultrasound, amniocentesis, and cordocentesis. Viral culture of neonatal urine can confirm the diagnosis after birth. Routine screening of the pregnant population is not currently recommended because present laboratory methods limit the ability to differentiate primary from recurrent infection. Preventive efforts at this time must focus on good hygiene, limited intimate contact with infected children, and responsible sexual practices.

Systematic review of the birth prevalence of congenital cytomegalovirus infection in developing countries.

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Congenital cytomegalovirus (CMV) infection is the leading infectious cause of congenital hearing loss and neurodevelopmental disability in developed countries. Information on congenital CMV infection in developing countries appears to be lacking. We conducted a systematic literature review to identify studies from developing countries with population-based samples of at least 300 infants that used laboratory methods established as reliable for the diagnosis of congenital CMV infection. Most studies were excluded due to biased samples or inadequate diagnostic methods; consequently the search identified just 11 studies that were from Africa, Asia, and Latin America. The number of newborns tested ranged from 317 to 12 195. Maternal CMV seroprevalence ranged from 84% to 100%. CMV birth prevalence varied from 0.6% to 6.1%. CMV-associated impairments were not documented in most studies. Birth prevalence ranges were higher than for Europe and North America, as expected based on the higher maternal CMV seroprevalence. With very limited data available on sequelae, the disease burden of congenital CMV in developing countries remains largely unknown at this time.

Risk Factors for Natural Hearing Evolution in Newborns With Congenital Cytomegalovirus Infection.

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Congenital cytomegalovirus (cCMV) is the major cause of congenital nonhereditary sensorineural hearing loss in children. Currently, criteria to identify infants at increased risk for unfavorable hearing outcome are lacking. To identify risk factors associated with cCMV-related hearing improvement, hearing deterioration, and late-onset hearing loss. This multicenter cohort study included patients from 6 secondary and tertiary hospitals enrolled in the Flemish CMV registry (Belgium). Newborns with untreated cCMV infection with at least 4-year audiological follow-up were included. Patients who presented with other possible causes of sensorineural hearing loss were excluded. Data were collected for 15 years (January 1, 2007, to February 7, 2022) and analyzed from September 26, 2022, to January 16, 2023. Primary outcome was hearing evolution (per-ear analysis; described as stable hearing, improvement, or deterioration). The association of gestational characteristics, clinical findings, timing of seroconversion, viral load, and hearing status at birth with hearing evolution was investigated using effect sizes (Cramer V, odds ratio [OR], or Hedges g). Of the 387 children, 205 of 385 with nonmissing data were male (53.2%), 113 (29.2%) had a symptomatic infection, and 274 (70.8%) had an asymptomatic infection. Every child was 4 years or older at final hearing evaluation. A total of 701 of 774 ears (90%) showed stable hearing (normal hearing or stable hearing loss since birth) over time. Late-onset hearing loss (normal hearing at birth followed by hearing loss) was present in 43 of 683 ears (6.3%). Among children with hearing loss present at birth, 24 of 34 ears (70.6%) had hearing deterioration, and 6 of 91 ears (6.6%) had hearing improvement. Prematurity was associated with a higher chance of hearing improvement (OR, 12.80; 95% CI, 2.03-80.68). Late-onset hearing loss was more prevalent in a first trimester infection (OR, 10.10; 95% CI, 2.90-34.48). None of the 104 ears of children with a third trimester seroconversion developed late-onset hearing loss. Findings of this cohort study support that ongoing audiological follow-up for untreated children with congenital hearing loss is important, as the majority of patients had hearing deterioration. The timing of seroconversion was associated with the risk of developing late-onset hearing loss. These insights can aid in parental counseling, patient stratification, and follow-up. Future research should focus on the effect of treatment, the influence of determined risk factors, and the study of eventual new risk factors in patients at high risk to develop hearing loss.

Cytomegalovirus (CMV) Infection Causes Degeneration of Cochlear Vasculature and Hearing Loss in a Mouse Model.

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Cytomegalovirus (CMV) infection is one of the most common causes of congenital hearing loss in children. We have used a murine model of CMV infection to reveal functional and structural cochlear pathogenesis. The cerebral cortex of Balb/c mice (*Mus musculus*) was inoculated with 2000 pfu (plaque forming units) of murine CMV on postnatal day 3. At 6 weeks of age, cochlear function was monitored using auditory brainstem response (ABR) and distortion product otoacoustic emission (DPOAE) measures. Histological assessment of cochlear vasculature using a corrosion cast technique was made at 8 weeks. Vascular casts of mCMV-damaged cochleas, and those of untreated control animals, were examined using scanning electron microscopy. We find very large variations in the degree of vascular damage in animals given identical viral injections (2000 pfu). The primary lesion caused by CMV infection is to the stria vascularis and to the adjacent spiral limbus capillary network. Capillary beds of the spiral ligament are generally less affected. The initial vascular damage is found in the mid-apical turn and appears to progress to more basal cochlear regions. After viral migration to the inner ear, the stria vascularis is the primary affected structure. We suggest that initial auditory threshold losses may relate to the poor development or maintenance of the endocochlear potential caused by stria dysfunction. Our increased understanding of the pathogenesis of CMV-related hearing loss is important for defining methods for

early detection and treatment.

The Effects of Viral Infections on the Molecular and Signaling Pathways Involved in the Development of the PAOs.

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Cytomegalovirus infection contributes to 10-30% of congenital hearing loss in children. Vertebrate peripheral auditory organs include the outer, middle, and inner ear. Their development is regulated by multiple signaling pathways. However, most ear diseases due to viral infections are due to congenital infections and reactivation and affect healthy adults to a lesser extent. This may be due to the fact that viral infections affect signaling pathways that are important for the development of peripheral hearing organs. Therefore, an in-depth understanding of the relationship between viral infections and the signaling pathways involved in the development of peripheral hearing organs is important for the prevention and treatment of ear diseases. In this review, we summarize the effects of viruses on signaling pathways and signaling molecules in the development of peripheral auditory organs.

Hearing loss in neonates and infants.

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Hearing in neonates and infants is crucial for their development of language and communication skills. Unless hearing loss is appropriately managed early, it can cause a significant socioeconomic burden considering its detrimental impact on the child's development and its common nature. It is also the most common congenital sensory deficit, with an approximate incidence of 1.5 per 1,000 newborns. Its etiologies are heterogeneous: genetic causes are reportedly involved in up to 80% of cases, while congenital cytomegalovirus infection is the leading environmental factor contributing to congenital hearing loss. The introduction of newborn hearing screening using automated auditory brainstem response and/or automated otoacoustic emission in many developed countries has helped detect and manage hearing loss early. Current auditory rehabilitation options such as cochlear implantation implementing cutting-edge technologies can treat almost all degrees of hearing loss, emphasizing the importance of early hearing detection and intervention. Rapidly developing genetic diagnostic technologies and future cutting-edge treatment options, including gene therapy, will shed light on the future management of hearing loss in neonates and infants.

Cytomegalovirus and the otolaryngologist.

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Human cytomegalovirus (CMV) is ubiquitous and has been documented as the etiologic factor in several diseases ranging from respiratory tract infection to congenital deafness. Special clinical manifestations of CMV of interest to the otolaryngologist are presented as related to the temporal bone and audiovestibular system, the facial nerve, the aerodigestive tract, the thyroid, the hematopoietic-lymphoreticular system, and salivary gland. In addition, current concepts of CMV oncogenic potential are discussed as they relate

to head and neck neoplasms. The results of a study performed on surgical pathology specimens obtained from 26 patients, in an attempt to isolate CMV in tissue culture from salivary gland and/or neoplasms of the head and neck, detected no evidence of viral presence. The significance of these findings and the role of future research is discussed.

Hearing Screening for Congenital CytoMegalovirus-Exploring Parents' Experiences of Completing Targeted Congenital Cytomegalovirus Screening at the Time of Their Infants' Newborn Hearing Screening.

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Background/Objectives: Congenital cytomegalovirus (cCMV) is the leading infectious cause of sensorineural hearing loss and neurodevelopmental disabilities, with prompt detection (<21 days of life) required to enable accurate diagnosis and anti-viral treatment where clinically appropriate. International guidelines recommend cCMV screening for infants who do not pass their Universal Newborn Hearing Screening (UNHS). This study aimed to explore parental experiences of targeted cCMV screening through the UNHS in Victoria, Australia between 2019 and 2020 (HearS-cCMV study). **Methods:** A qualitative study comprising 18 semi-structured interviews with parents who took saliva swabs from their infants who did not pass their UNHS. A maximum variation sampling strategy was used with data analysed using thematic analysis. **Results:** Four themes described 18 parents' experiences of cCMV screening: (1) parents' lack of CMV awareness prior to cCMV screening; (2) overall positive experience; (3) varied understanding of CMV post screening; and (4) parents were glad to screen their infant for cCMV. Enablers of targeted cCMV screening included the swab being simple and non-invasive, being easier to complete in the hospital than at home, and the screening being well delivered by the staff. Barriers included a potential increase in anxiety, especially with false positives, and the timing of cCMV screening coinciding with their infant not passing UNHS being difficult for some parents. **Conclusions:** Parent experiences of targeted cCMV screening were positive. Increasing public knowledge of cCMV and training staff members to complete the CMV swab would reduce the risk of false positives and associated parental anxiety. This would facilitate successful routine targeted cCMV screening.
